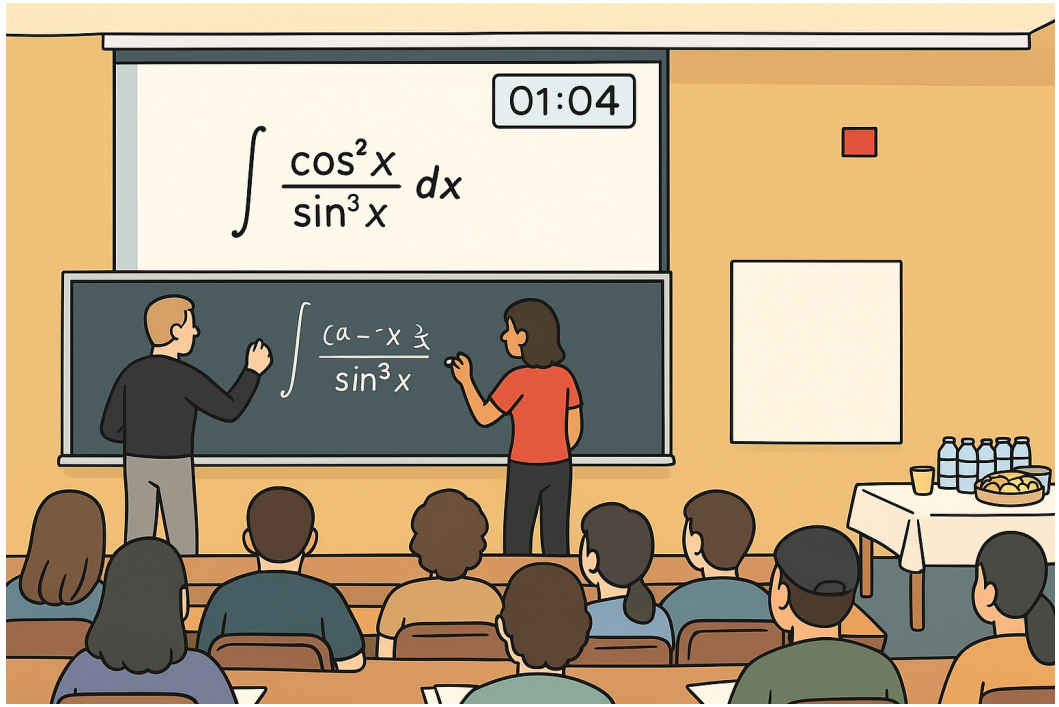
Integratión Bee Physics
Semifinal



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

Semifinal

GRUPO 1

Integral #56 **Integratión Bee Physic**

$$\int_2^{\infty} \frac{\lfloor x \rfloor \, x}{x^4 - 1} \, \mathrm{d}x$$

Respuesta Integral #56

Integratión Bee Physic

$$\int_2^{\infty} \frac{\lfloor x \rfloor \ x}{x^4 - 1} \mathrm{d}x =$$

$$\sum_{n=2}^{\infty} \frac{n}{4} \left(\ln \left| \frac{(n+1)^2 - 1}{(n+1)^2 + 1} \right| - \ln \left| \frac{n^2 - 1}{n^2 + 1} \right| \right)$$

Integral #74 **Integratión Bee Physic**

$$\int \frac{1}{\sqrt{(x-a)(b-x)}} \mathrm{d}x$$

$$\int \frac{1}{\sqrt{(x-a)(b-x)}}\,\mathrm{d}x =$$

$$2\sin^{-1}\sqrt{\frac{x-a}{b-a}} + C$$

Integral #54 **Integratión Bee Physic**

$$\int \frac{1}{\sin x + \sin 2x} \, \mathrm{d}x$$

Respuesta Integral #54

Integratión Bee Physic

$$\int \frac{1}{\sin x + \sin 2x}\,\mathrm{d}x =$$

$$\frac{1}{2}\ln|1+\cos x| + \frac{1}{6}\ln|1-\cos x| - \frac{2}{3}\ln|1+2\cos x| + C$$